

UNIVERSITY OF AMSTERDAM

MASTER'S THESIS

Interventions in Asymmetric Belief Systems

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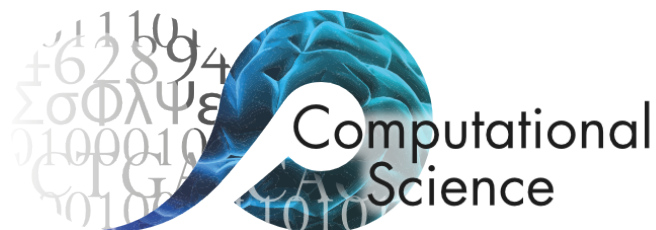
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for the degree of Master of Science in Computational Science*

in the

Computational Science Lab
Informatics Institute

June 2026



Declaration of Authorship

I, Henry ZWART, declare that this thesis, entitled ‘Interventions in Asymmetric Belief Systems’ and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at the University of Amsterdam.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

A handwritten signature in black ink, appearing to read 'Henry Zwart', with a stylized flourish at the end.

Date: 29 June 2026

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Abstract

Faculty of Science
Informatics Institute

Master of Science in Computational Science

Interventions in Asymmetric Belief Systems

by Henry ZWART

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Thank the people that have helped: supervisors, family, etc.

Use of AI

Statement that generative AI was not used in the writing of this thesis.

Contents

Declaration of Authorship	i
Abstract	ii
Acknowledgements	iii
Use of AI	iv
Contents	v
List of Figures	vii
List of Tables	viii
Abbreviations	ix
1 Terminology and notation	1
2 Introduction	2
2.1 Plan	2
2.2 Research questions	2
2.3 Proposed thesis structure	3
3 Asymmetric belief systems	6
3.1 Plan	6
3.2 Belief system dynamics	6
3.3 Non-equilibrium belief system model	7
3.4 Simulation via Glauber dynamics	10
3.5 Modelling interventions	10
4 Methods	11
4.1 Plan	11
4.2 Parameter estimation	11
4.2.1 Regularisation	14

5	Existence and impact of asymmetry in belief systems	18
5.1	Plan	18
5.2	Fitting model	19
5.3	RQ2.1: Asymmetry in estimated models	20
5.4	RQ3.1: Intervention strategy and effectiveness under sym/asym assumptions	21
5.4.1	Ranking interventions	22
5.4.2	Intervention effect	23
6	Heterogeneous belief systems and intervention effects	25
6.1	Plan	25
6.2	RQ2.2: differences between belief systems for different (types of) individuals	26
6.3	RQ3.2: Intervention strategy and effectiveness across individuals	26
6.3.1	Intervention ranking	26
7	Discussion	28
7.1	Plan	28
8	Climate beliefs dataset	29
8.1	Plan	29
8.2	Longitudinal climate attitudes survey	29
8.3	Question selection	31
8.4	Validation	31
8.5	Normalisation, cleaning, and transformations	31
8.6	Data Validation	32
9	Related work	36
10	Conclusions and future work	37
11	Ethics and Data Management	39
	Bibliography	40

List of Figures

Figure 4.1	Effect of regularisation on model sparsity	16
Figure 4.2	Regularisation strength EBIC	17
Figure 5.1	Climate belief system network (asymmetric)	19
Figure 5.2	Climate belief system interaction effect matrix	20
Figure 5.3	Directional interaction differentials.	21
Figure 5.4	Intervention ranking targeting climate policy support (weak intervention)	22
Figure 5.5	Intervention ranking targeting climate policy support (medium intervention)	22
Figure 5.6	Intervention ranking targeting climate policy support (perfect intervention)	23
Figure 5.7	Effect of weak interventions targeting climate policy support	23
Figure 5.8	Effect of medium interventions targeting climate policy support	24
Figure 5.9	Effect of perfect interventions targeting climate policy support	24
Figure 6.1	<i>Conservative</i> and <i>Liberal</i> belief networks	26
Figure 6.2	Ranked intervention effects per-individual, weak, targeting climate policy	27
Figure 6.3	Ranked intervention effects per-individual, perfect, targeting climate policy	27
Figure 8.1	Longitudinal survey response dates per-wave	30
Figure 8.2	Between-response time distribution	30

List of Tables

Table 8.1	Dataset type coercion mapping
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33

Abbreviations

CSL Computational Science **L**ab

UvA Universiteit van **A**msterdam

Chapter 1

Terminology and notation

Define key terminology, outline notation used in following sections.

Chapter 2

Introduction

2.1 Plan

- Motivate the problem
- Contributions
- Research questions (thinking these are perhaps better left to later, in favour of contributions here)

2.2 Research questions

Theoretical contributions:

- **RF1.1:** Extending the causal attitude network model of belief systems to: (i) support asymmetric causal effects between beliefs and attitudes, and (ii) model intervention dynamics.

Inferring belief systems:

- **RQ2.1:** To what extent are causal relations *symmetric* or *asymmetric* in models of climate change belief systems inferred from observational data?
- **RQ2.2:** How do (symmetric or asymmetric) belief systems relating to climate change vary between conservative and liberal individuals?

Intervention dynamics:

- **RQ3.1:** How do asymmetric and symmetric beliefs systems inferred from the climate attitudes dataset differ with regards to intervention strategy and effectiveness?

- **RQ3.2:** How do intervention outcome and effectiveness vary between individuals with different initial conditions, or between conservative and liberal individuals?

2.3 Proposed thesis structure

- **Terminology and notation**
- **Introduction:**
 - Motivate the problem,
 - Outline contributions (research questions)
- **Asymmetric belief system:**
 - Define and illustrate the model
 - How do we model interventions?
- **Methods:**
 - Model simulation with Glauber dynamics
 - Counterfactual intervention experiments — comparing against the no-intervention scenario, measuring differences in effects.
 - Parameter estimation:
 - Maximum likelihood estimation
 - Conditional on a specific binarisation
 - Marginalising over binarisation process
 - Regularisation
- **Existence and impacts of asymmetry in belief systems**
 - Results for:
 - **RF1:** Show asymmetric model fit
 - **RQ2.1:** Existence of asymmetric relations. Some, but not all, are significant. Categorising relations into types: symmetric, asymmetric (both directions exist, with different effect sizes), and unidirectional (only one direction exists).
 - **RQ3.1:** Differences between symmetric and asymmetric models, with regards to intervention strategy (which intervention to do) and effectiveness (magnitude of change compared to the no-intervention case).
- **Individual heterogeneity in belief systems and intervention dynamics**
 - Second results section
 - **RQ2.2:** Fit models to conservative and liberal subsets of the data; compare and contrast. Compare with the model from the previous section, i.e., fit on the entire dataset.

► **RQ3.2:**

- Distribution of intervention effectiveness by individual.
- How does intervention ranking vary across individuals?
- Characterising how different initial states affect intervention success.
- Looking at how other theory-driven features affect success, e.g.:
 - How receptive is the individual to the intervention?

The effective baseline ($h_i + \sum_j J_{ji}s_j$) determines the probability that $S_i^{t+1} = s$ for a state $s \in \pm 1$. Evaluating this after intervening provides a measurement for the success of the intervention on the intervention spin itself.

We could look at how this changes with different intervention strengths (it follows a logistic curve).

- How consistent is the individual's belief state, as measured by the total system energy?
- How 'entrenched' is the target attitude?

Measure $h_k s_k + \sum_j J_{jk} s_j s_k$, where k is the target attitude.

• **General discussion**

- Draw out the key findings from the above two results sections
- Make clear the implications for our theory of belief system dynamics
- Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
- Representational limitations of our model:
 - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can't capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
 - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as 'on/off', where the 'off' state has *no* effect on other spins, rather than a negative effect.
 - More generally, effect sizes may vary depending on specific spin states.

• **Dataset**

- Introduce the (insert name here) dataset, give context, survey details
- Validation, cleaning, transformations
- Question selection, indexes

- Binarisation
- **Literature review/related work**
- **Conclusions and future work**

Chapter 3

Asymmetric belief systems

3.1 Plan

- Define and illustrate the model
- Model simulation with Glauber dynamics
- How do we model interventions?

3.2 Belief system dynamics

We will now formalise the dynamics of an individual belief system, under the following assumptions:

1. Beliefs and attitudes can be treated as discrete entities.
2. The state of a belief or attitude is Markovian with respect to the previous system configurations.
3. The probability distribution relating current and future belief states does not change over time.

The first assumption includes two key points. Firstly, that it makes sense to consider beliefs and attitudes as *entities*, or *objects*. This contrasts, for instance, alternative interpretations of attitudes as emissions from an underlying structure. (**TODO:** review CAN paper for further discussion on this matter). Secondly, that beliefs and attitudes can be considered *discrete* or *separable*. This is to say that there are no absolute physical constraints on the combinations of states which may be observed (although certain

combinations may be highly unlikely). In particular, this captures the requirement that each belief or attitude can be represented using a single state variable.

Our second assumption concerns the treatment of beliefs and attitudes as Markovian.

- Harder to justify conceptually, since individuals have memory and can recall prior states beyond the previous second.

Consider a belief or attitude S_i with domain Ω_{S_i} . The state of S_i at time t depends on the previous states of other beliefs and attitudes, and we can therefore describe the trajectory of S_i as a Markov process:

$$\{s_i^t\}_{t=1}^{\infty} \quad \text{where} \quad s_i^t \in \Omega_{S_i} \quad (3.1)$$

with conditional state probability $P(S_i^{\tau+1} \mid \mathbf{S}^1, \dots, \mathbf{S}^{\tau}) = P(S_i^{\tau+1} \mid \mathbf{S}^{\tau})$ for $\tau \in \mathbb{N}$.

Since the state of S_i^{t+1} depends only on the previous belief system state, it follows that for any other belief or attitude S_j , the states S_i^{t+1} and S_j^{t+1} are conditionally independent given $\mathbf{S}^t$. We can therefore straightforwardly extend Equation 3.1 to describe the evolution of the entire belief system as a Markov process,

$$\{\mathbf{s}^t\}_{t=1}^{\infty} \quad \text{where} \quad \mathbf{s}^t \in \Omega_{S_1} \times \dots \times \Omega_{S_n} \quad (3.2)$$

with the probability of each belief system configuration given by the conditional distribution $P(\mathbf{S}^{t+1} \mid \mathbf{S}^t)$.

Under this conceptualisation, the task of modelling a belief system thus reduces to describing how the instantaneous configuration of belief states affects the probability distribution over possible future states.

3.3 Non-equilibrium belief system model

When studying the *symmetric* Ising model it is common to assume that the model is at equilibrium, such that $P(\mathbf{S}^t = \mathbf{s} \mid \mathbf{S}^{t-1}) = P(\mathbf{S} = \mathbf{s})$ for all configurations $\mathbf{s}$ and observation times t . The probability of observing $\mathbf{s}$ is then time-invariant and given by the Boltzmann probability parameterised by the Hamiltonian $H(\mathbf{s})$ (Cardy, 1996; Christensen & Moloney, 2005):

$$P(\mathbf{S} = \mathbf{s}) = \frac{e^{-\frac{1}{k_b T} \cdot H(\mathbf{s})}}{\sum_{\mathbf{s}'} e^{-\frac{1}{k_b T} \cdot H(\mathbf{s}')}} \quad (3.3)$$

where k_b is the Boltzmann constant, and the temperature parameter $T \in \mathbb{R}^+$ controls stochasticity. Equation 3.3 converges to a uniform distribution over all configurations when $T \rightarrow +\infty$, and a uniform distribution over the restricted set of configurations for which $H(\mathbf{s}')$ is minimised when $T \rightarrow 0^+$.

For the purposes of this study, we are interested in intervention dynamics, which are inherently non-equilibrium. To see why this is the case, notice that the purpose of intervention is to change the distribution of observed states. This is true both interventions intended to change the dominant state (e.g., to promote a sustainable alternative to a typical unsustainable behaviour) or reinforce it. Any belief system model intended for studying intervention dynamics therefore cannot assume equilibrium, and cannot define model dynamics using the Boltzmann distribution.

We instead define the conditional distribution $P(\mathbf{S}^t = \mathbf{s} \mid \mathbf{S}^{t-1})$ explicitly. Recall that the states of each pair of spins S_i, S_j at time t are conditionally independent random variables, given knowledge of the previous configuration $\mathbf{S}^{t-1}$ (Chapter 3.2). It therefore suffices for us to describe the distribution over spin states for an individual spin S_i at time t , as conditional on $\mathbf{S}^{t-1}$.

As in the standard Ising model formulation, we will assume that the model typically evolves in the direction of lower energy; however, rather than defining the energy at the level of the model using the Hamiltonian, we consider the energy experienced by S_i as a result of its baseline activation (also known as a *local field effect*), and interactions with other spins.

The baseline activation $h_i \in \mathbb{R}$ describes the tendency for S_i to take on the value $+1$ in the absence of interactions with other spins, or, more generally, when pairwise spin interactions have a net-effect of zero. S_i tends to adopt the value $+1$ under these circumstances, iff, $h_i > 0$, and -1 iff $h_i < 0$. This tendency increases with the magnitude of h_i .

Other spins influence S_i 's state through alignment or opposition relations. For a spin S_j , we define an interaction effect J_{ji} (read '*influence of j on i* '). When J_{ij} is positive or negative, S_i is influenced to adopt the state S_j^{t-1} or $-S_j^{t-1}$ respectively. In the special case when $j = i$, we refer to J_{ii} as a *self-influence* effect. Positive self-influence effects reflect the inertia of S_i , i.e., the tendency to sustain a particular belief or attitude irrespective of other beliefs and attitudes. Negative self-influence effects are not clearly interpretable. The inclusion of self-influence effects allows us to capture the timescales of different beliefs or attitudes independently.

The energy experienced by S_i for a particular spin state $s \in \{0, 1\}$ is then the result of s in combination with the baseline activation and influence effects from all spins with edges to S_i :

$$H_i(s \mid \mathbf{s}^{t-1}) = h_i \cdot s + \sum_j A_{ji} J_{ji} s_j^{t-1} \cdot s \quad (3.4)$$

where $\mathbf{A} \in \{0, 1\}^{n \times n}$ is a pairwise adjacency matrix, with $A_{ji} = 1$, iff, S_j influences S_i . Dividing the common factor of s , we can equivalently write Equation 3.4 as

$$H_i(s \mid \mathbf{s}^{t-1}) = s \cdot h_i^{\text{eff}}(\mathbf{s}^{t-1}) \quad (3.5)$$

where $h_i^{\text{eff}}(\mathbf{s}^{t-1}) = h_i + \sum_j A_{ji} J_{ji} s_j^{t-1}$ is the effective baseline activation experienced by S_i at time t . We then define the probability that S_i adopts the state s at time t as:

$$\begin{aligned} P(S_i^t = s \mid \mathbf{S}^{t-1} = \mathbf{s}) &= \frac{e^{\frac{1}{T} H_i(s \mid \mathbf{s})}}{\sum_{s' \in \{0, 1\}} e^{\frac{1}{T} H_i(s' \mid \mathbf{s})}} \\ &= \frac{e^{\frac{1}{T} H_i(s \mid \mathbf{s})}}{e^{\frac{1}{T} H_i(s \mid \mathbf{s})} + e^{-\frac{1}{T} H_i(-s \mid \mathbf{s})}} \\ &= \frac{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\mathbf{s})}}{e^{\frac{1}{T} s \cdot h_i^{\text{eff}}(\mathbf{s})} + e^{-\frac{1}{T} s \cdot h_i^{\text{eff}}(\mathbf{s})}} \end{aligned} \quad (3.6)$$

For $s = +1$, this reduces to the logistic function:

$$\begin{aligned} p_i^s &:= P(S_i^t = +1 \mid \mathbf{S}^{t-1} = \mathbf{s}) = \frac{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})}}{e^{\frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})} + e^{-\frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})}} \\ &= \frac{1}{1 + e^{-2 \cdot \frac{1}{T} \cdot h_i^{\text{eff}}(\mathbf{s})}} \\ &= \text{logistic}(2h_i^{\text{eff}}(\mathbf{s})/T) \end{aligned} \quad (3.7)$$

Therefore the complete conditional distribution is given by:

$$\begin{aligned} P(\mathbf{S}^t = \mathbf{s}^t \mid \mathbf{S}^{t-1} = \mathbf{s}^{t-1}) &= \prod_{i=1}^n P(S_i^t = s_i \mid \mathbf{S}^{t-1} = \mathbf{s}^{t-1}) \\ &= \prod_{i=1}^n \left[\left(\frac{1 + s_i}{2} \right) p_i^{s^{t-1}} + \left(\frac{1 - s_i}{2} \right) (1 - p_i^{s^{t-1}}) \right] \end{aligned} \quad (3.8)$$

NOTE: The initial distribution must be specified, since the model only captures transition probabilities.

3.4 Simulation via Glauber dynamics

3.5 Modelling interventions

Chapter 4

Methods

4.1 Plan

- Counterfactual intervention experiments — comparing against the no-intervention scenario, measuring differences in effects.
- Parameter estimation via MLE:
 - Marginalising over binarisation process
 - Regularisation

4.2 Parameter estimation

Rough plan:

1. We estimate parameters using maximum likelihood estimation.
2. Naïve maximum likelihood estimation identifies parameterisation θ^* such that

$$\theta^* = \operatorname{argmax}_{\theta \in \mathbb{R}^p} \mathcal{L}_{D_B}(\hat{\theta}) \quad (4.1)$$

for a binarisation $D_B \sim \text{Bin}(D)$, where $\mathcal{L}_{D_B}(\hat{\theta}) = \log P(D_B \mid \hat{\theta})$. This is subject to two sources of uncertainty: (i) due to sample size, and (ii) due to fitting to the particular binarisation D_B .

3. We introduce measures to counteract each of these forms of uncertainty. First, we apply regularisation to counteract sample size uncertainty. Second, we marginalise over all possible binarisations of the dataset.

4. We instead choose θ^* such that

$$\theta^* = \operatorname{argmax} \left\{ \mathbb{E}_{D_B} [\mathcal{L}_{D_B}(\hat{\theta}) \mid \mathcal{D} = D] - \sum_{\theta \in \hat{\theta}} \sqrt{\theta^2 + \varepsilon} \quad : \quad \hat{\theta} \in \mathbb{R}^p \right\} \quad (4.2)$$

The first term is the expected log-likelihood of a binarised dataset given the candidate parameterisation $\hat{\theta}$:

$$\mathbb{E}_{D_B} [\mathcal{L}_{D_B}(\hat{\theta}) \mid \mathcal{D} = D] = \sum_{D_B} P(\operatorname{Bin}(D) = D_B \mid \mathcal{D} = D) \cdot \mathcal{L}_{D_B}(\hat{\theta}) \quad (4.3)$$

and the second term is a variation on L1 regularisation, for which the first derivative is continuous, and thus amenable to gradient-based optimisation. We choose ε small, such that this remains a good approximation to the standard formulation of L1 regularisation.

Note: this section was written prior to the above plan. I think it spends too much time on the non-marginalised log-likelihood derivation, so I'm considering moving this to the appendix, and keeping this section short.

For a collection of observations D drawn from a system assumed to be described by a model $\mathcal{M}$ with $p \in \mathbb{N}$ parameters, but for which the true parameter values are unknown, parameter estimation is the task of inferring the parameterisation responsible for generating D . For the purposes of this study we use maximum likelihood estimation to identify the belief system model parameterisation which maximises the likelihood of the observed data.

First, let us formally outline the setting. We consider a population of $M \in \mathbb{N}$ individuals, assumed to share a common belief system $\mathcal{M}$ comprising $N \in \mathbb{N}$ spins (beliefs and attitudes). Suppose that for each individual we have observed the (possibly continuous) states of their beliefs and attitudes at each of $Q \in \mathbb{N}$ uniformly-spaced points in time:

$$D = \left[\{\mathbf{x}_1^t\}_{t=1}^Q, \dots, \{\mathbf{x}_M^t\}_{t=1}^Q \right] \quad (4.4)$$

from which we have sampled a binarised dataset:

$$D_B = \left[\{\mathbf{s}_1^t\}_{t=1}^Q, \dots, \{\mathbf{s}_M^t\}_{t=1}^Q \right] \sim \operatorname{Binarise}(D) \quad (4.5)$$

For the time being will consider the task of parameter estimation with respect to the particular binarisation D_B , but will return to the more general problem of estimating parameters with respect to D itself in (reference section on marginalised MLE optimisation).

Given a parameterisation $\hat{\theta} \in \mathbb{R}^p$ for $p \in \mathbb{N}$ model parameters, the *likelihood* of D_B given $\hat{\theta}$ is:

$$\begin{aligned}
L_{D_B}(\hat{\theta}) &= P(\mathcal{D} = D_B \mid \Theta = \hat{\theta}) \\
&= \prod_{m=1}^M P_{\{s_m^t\}|\Theta}(\{s_m^t\}_{t=1}^Q \mid \hat{\theta}) \\
&= \prod_{m=1}^M P_{S_m^1 \dots S_m^Q|\Theta}(s_m^1, \dots, s_m^Q \mid \hat{\theta}) \\
&= \prod_{m=1}^M \prod_{t=1}^Q P_{S_m^t|S_m^1 \dots S_m^{t-1}\Theta}(s_m^t \mid s_m^1, \dots, s_m^{t-1}, \hat{\theta}) \tag{4.6} \\
&= \prod_{m=1}^M P_{S_m^1|\Theta}(s_m^1 \mid \hat{\theta}) \prod_{t=1}^Q P_{S_m^t|S_m^{t-1}\Theta}(s_m^t \mid s_m^{t-1}, \hat{\theta}) \quad (*) \\
&= P_0(D_B) \prod_{m=1}^M \prod_{t=1}^Q P_{S_m^t|S_m^{t-1}\Theta}(s_m^t \mid s_m^{t-1}, \hat{\theta})
\end{aligned}$$

where $P_0(D_B) = \prod_{m=1}^M P(S_m^1 = s_m^1)$ is a pre-specified initial condition which is independent of the parameterisation, and $(*)$ follows from the Markov assumption in our belief system model.

Maximum likelihood estimation identifies the parameterisation $\theta^* \in \mathbb{R}^p$ which maximises the likelihood. In practice, however, to avoid potential numerical instability in evaluating $L_{D_B}(\hat{\theta})$, arising from the multiplication of many small conditional probabilities, we instead maximise the *log-likelihood*:

$$\theta^* = \underset{\theta \in \mathbb{R}^p}{\operatorname{argmax}} \mathcal{L}_{D_B}(\hat{\theta}) \tag{4.7}$$

Where $\mathcal{L}_{D_B}(\hat{\theta}) = \log L_{D_B}(\hat{\theta})$ is given by

$$\mathcal{L}_{D_B}(\hat{\theta}) = \sum_{m=1}^M \sum_{t=1}^Q \log P(S_m^t = s_m^t \mid S_m^{t-1} = s_m^{t-1}, \Theta = \hat{\theta}) \tag{4.8}$$

We omit $P_0(D_B)$ from Equation 4.8 since it is independent of $\hat{\theta}$ and therefore does not affect the value of θ^* .

Substituting the conditional transition probability from Equation 3.8, we obtain a concrete form for the log-likelihood:

$$\begin{aligned}
\mathcal{L}_{D_B}(\hat{\theta}) &= \sum_{m=1}^M \sum_{t=1}^Q \log \left(\prod_{i=1}^N P_{\Theta=\hat{\theta}}(S_{i,(m)}^t = s_{i,(m)}^t \mid S_{i,(m)}^t = s_{i,(m)}^t) \right) \\
&= \sum_{m=1}^M \sum_{t=1}^Q \sum_{i=1}^N \log P_{\Theta=\hat{\theta}}(S_{i,(m)}^t = s_{i,(m)}^t \mid S_{i,(m)}^t = s_{i,(m)}^t) \\
&= \sum_{m=1}^M \sum_{t=1}^Q \sum_{i=1}^N \log \left[\frac{\exp(s_{i,(m)}^t \cdot h_i^{\text{eff}}(s_m^{t-1}))}{\exp(h_i^{\text{eff}}(s_m^{t-1})) + \exp(-h_i^{\text{eff}}(s_m^{t-1}))} \right] \quad (4.9) \\
&= \sum_{m=1}^M \sum_{t=1}^Q \sum_{i=1}^N \log \left[\frac{\exp(s_{i,(m)}^t \cdot h_i^{\text{eff}}(s_m^{t-1}))}{2 \cosh(h_i^{\text{eff}}(s_m^{t-1}))} \right] \\
&= \sum_{m=1}^M \sum_{t=1}^Q \sum_{i=1}^N s_{i,(m)}^t \cdot h_i^{\text{eff}}(s_m^{t-1}) - \log(2 \cosh h_i^{\text{eff}}(s_m^{t-1}))
\end{aligned}$$

where $h_i^{\text{eff}}(\mathbf{s})$ is the effective baseline activation (Equation 3.5), and the reduction of the denominator follows from the identity $\cosh(x) = \frac{e^x + e^{-x}}{2}$, as also used in (Nguyen et al., 2017).

Equation 4.7 is maximised when $\frac{\partial \mathcal{L}_{D_B}}{\partial \theta_i}(\hat{\theta}) = 0$ for each parameter θ_i . Recall that the model defined in Chapter 3.3 includes two kinds of parameters: baseline activations h_i and influence effects J_{ji} , such that $\hat{\theta} = \langle \mathbf{h}, \mathbf{J} \rangle$. The partial derivative of the log-likelihood with respect to an arbitrary parameter θ in $\mathbf{h}$ or $\mathbf{J}$ is:

$$\frac{\partial \mathcal{L}_{D_B}(\hat{\theta})}{\partial \theta} = \sum_{m=1, t=1, n=1}^{M, Q, N} \dots \quad (4.10)$$

4.2.1 Regularisation

Both the symmetric and asymmetric belief system models contain a relatively large number of parameters to be estimated compared to the number of observations in the dataset, which poses an issue for obtaining reliable parameter estimates (Epskamp et al., 2018). In line with standard practice, we apply regularisation during estimation to encourage model sparsity, using a variation of LASSO regularisation (Tibshirani, 1996) for which the first derivative is smooth.

LASSO regularisation (least absolute shrinkage and selection operator) is implemented as an additional summand in the objective function. Rather than maximising the log-likelihood directly, we choose parameters $\hat{\boldsymbol{\theta}} \in \mathbb{R}^p$ such that

$$\boldsymbol{\theta}^* = \operatorname{argmax}_{\boldsymbol{\theta} \in \mathbb{R}^p} \left\{ \mathcal{L}_D(\hat{\boldsymbol{\theta}}) - \lambda \sum_{i=1}^p |\hat{\theta}_i| \right\} \quad (4.11)$$

This has the effect of penalising non-zero parameters with negligible contribution to the log-likelihood. The parameter $\lambda \in \mathbb{R}^+$ controls regularisation strength, with larger values resulting in sparser models.

This formulation of LASSO regularisation has a discontinuous first-derivative at $\hat{\theta}_i = 0$ due to the absolute value function. Instead, we use the following smooth variation, where $\varepsilon \in \mathbb{R}^+$:

$$\boldsymbol{\theta}^* = \operatorname{argmax}_{\boldsymbol{\theta} \in \mathbb{R}^p} \left\{ \mathcal{L}_D(\hat{\boldsymbol{\theta}}) - \lambda \sum_{i=1}^p \sqrt{\hat{\theta}_i^2 + \varepsilon} \right\} \quad (4.12)$$

Equation 4.12 converges to Equation 4.11 as $\varepsilon \rightarrow 0^+$, and has a well-defined first-derivative for $\varepsilon > 0$, namely:

$$\frac{d}{d\hat{\theta}_i} \left(-\lambda \sum_{j=1}^p \sqrt{\hat{\theta}_j^2 + \varepsilon} \right) = \frac{\lambda \hat{\theta}_i}{\sqrt{\hat{\theta}_i^2 + \varepsilon}} \quad (4.13)$$

For the purposes of this study, we will consider parameters as ‘effectively non-zero’ if their magnitude exceeds 10^{-2} . We thus choose $\varepsilon = 10^{-8}$, such that $\sqrt{\varepsilon}$ is two orders of magnitude smaller than the minimal parameter size of interest, ensuring that Equation 4.12 is a reasonable approximation to Equation 4.11 for parameter values smaller than 10^{-2} .

- Effects: How does regularisation affect sparsity? (In progress; planning a different figure)

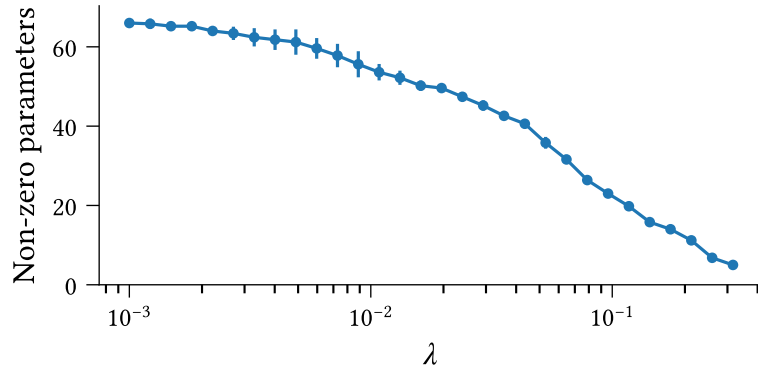


Figure 4.1: Model sparsity increases with regularisation strength (λ). Error bars display 95% confidence intervals around expected number of non-zero parameters, measured across different stochastic binarisations of observed data.

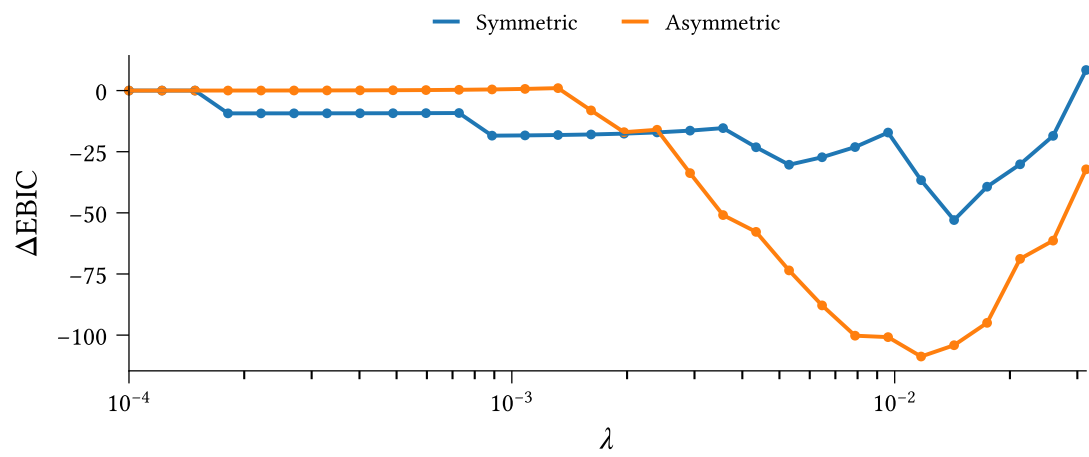
We use the Extended Bayesian Information Criterion (EBIC, Equation 4.14) (Chen & Chen, 2008) to select an appropriate regularisation strength, as recommended in Epskamp et al. (2018). The EBIC is an evaluative criterion for model selection, which balances maximisation of the log-likelihood with model complexity, as measured by the number of parameters. Compared to the original Bayesian Information Criterion, EBIC tends to be more conservative when the number of parameters and observations are comparable (Foygel & Drton, 2010).

For a model with p parameters, k of which are non-zero, fit using a dataset with n observations, the EBIC is defined as:

$$\text{EBIC}(\boldsymbol{\theta}^*) = k \cdot [\ln(n) + 2\gamma \ln(p)] - 2\mathcal{L}_D(\boldsymbol{\theta}^*) \quad (4.14)$$

We take $\gamma = 0.25$, which has been shown to work well for network inference in the inverse Ising problem (Barber & Drton, 2015).

In general, models with a smaller EBIC are preferred, as this indicates a reasonable trade-off between model fit and complexity. We fit each of the symmetric and asymmetric models for candidate regularisation strengths $\lambda \in \mathbb{R} \subset [10^{-4}, 10^{-1}]$, and select the values for which the resulting EBIC is minimal (Figure 4.2). This yields symmetric and asymmetric regularisation strengths of 0.014 and 0.012 respectively.

Figure 4.2: **TODO**

Chapter 5

Existence and impact of asymmetry in belief systems

5.1 Plan

First results section, for experiments on the effect of asymmetry

- **RF1:** Show asymmetric model fit
- **RQ2.1:** Existence of asymmetric relations. Some, but not all, are significant. Categorising relations into types: symmetric, asymmetric (both directions exist, with different effect sizes), and unidirectional (only one direction exists).
- **RQ3.1:** Differences between symmetric and asymmetric models, with regards to intervention strategy (which intervention to do) and effectiveness (magnitude of change compared to the no-intervention case).

5.2 Fitting model

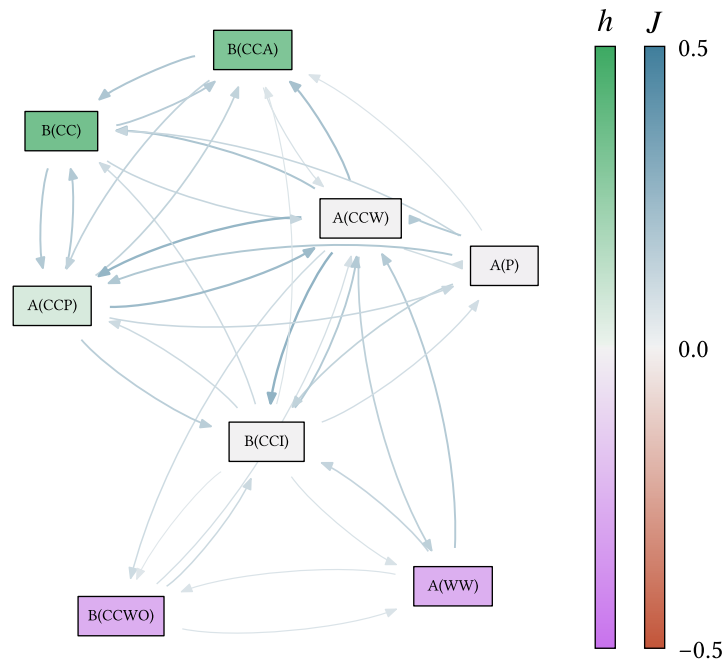


Figure 5.1: TODO

5.3 RQ2.1: Asymmetry in estimated models

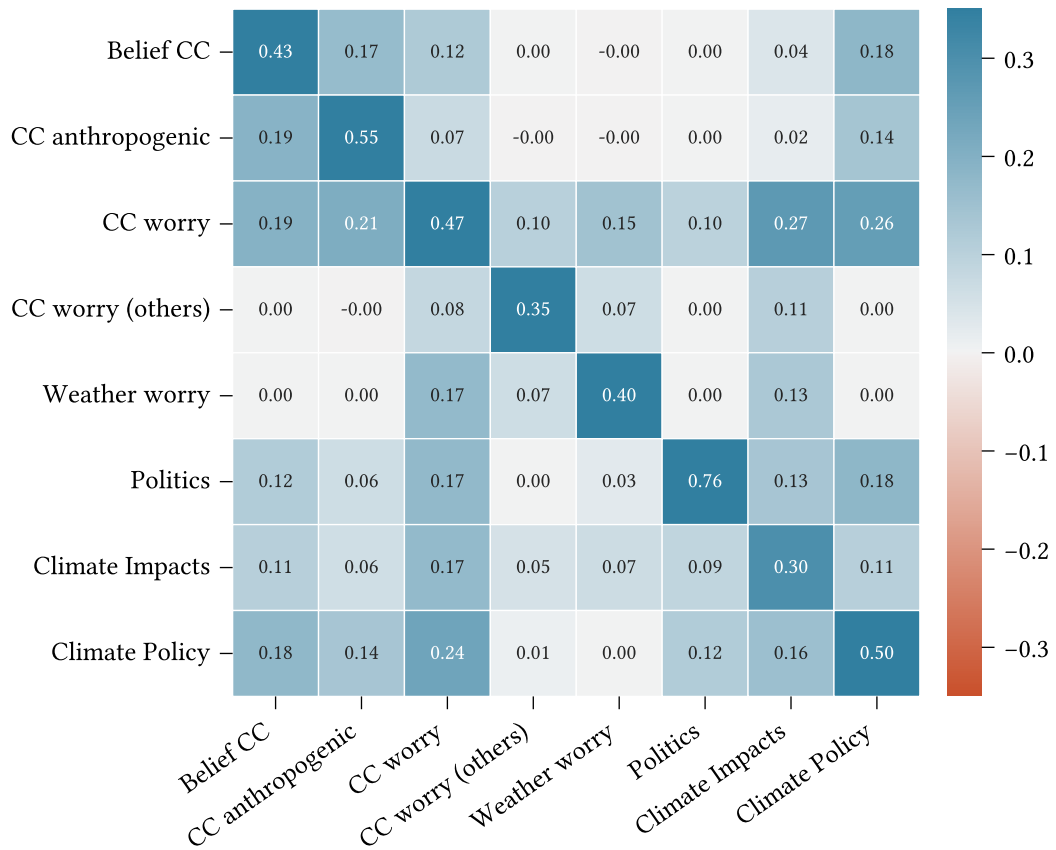


Figure 5.2: Climate belief system interaction effect matrix.

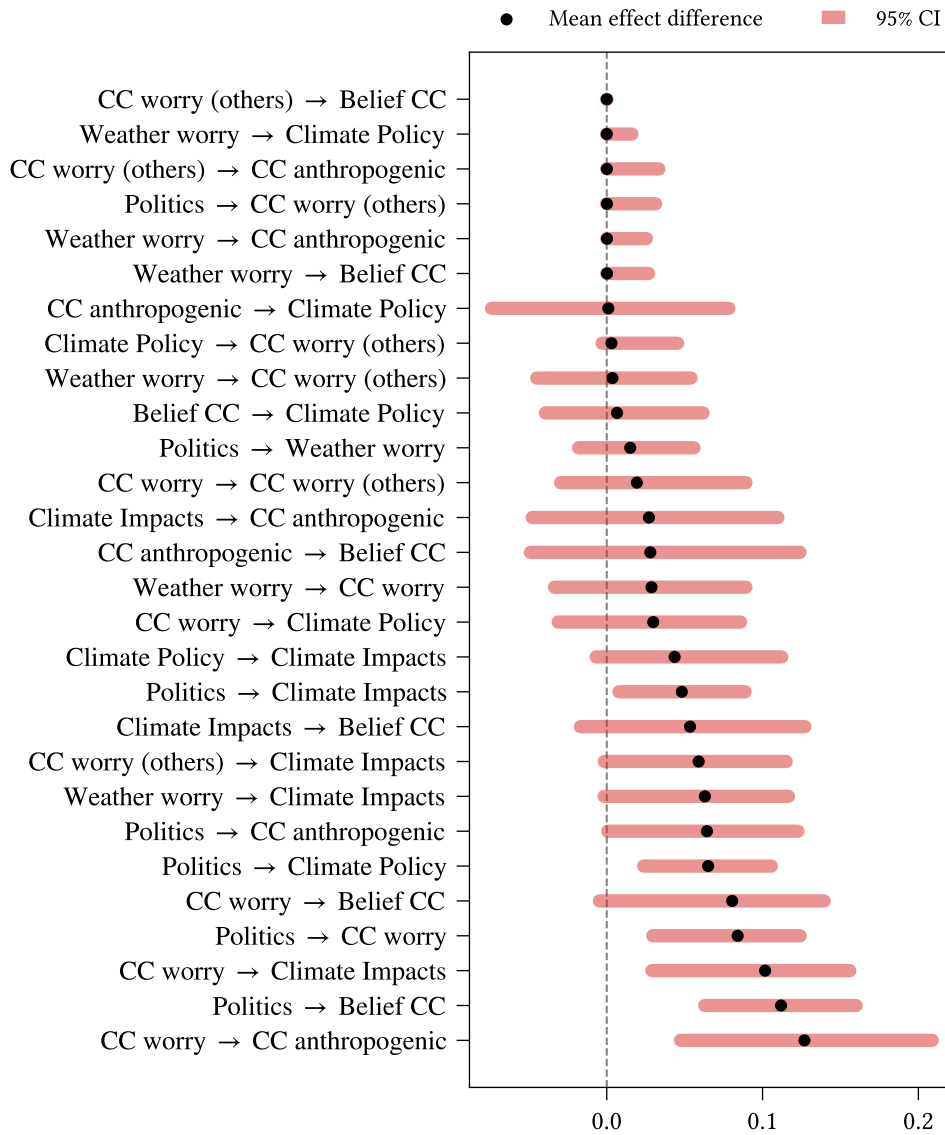


Figure 5.3: TODO

5.4 RQ3.1: Intervention strategy and effectiveness under sym/asym assumptions

Fit models using bootstrapping. Consider both effects of intervention (how does intervening at S affect other spins) and strategies (how does intervening at different spins affect this *one*). Repeat for varying levels/strengths of intervention.

For strategy, estimate expected ranking of interventions at both collective and individual levels. For the former, measure collective effect, then rank interventions, then take mean

across repeats. For the latter, measure expected individual effect, rank interventions, then take mean across individuals.

5.4.1 Ranking interventions

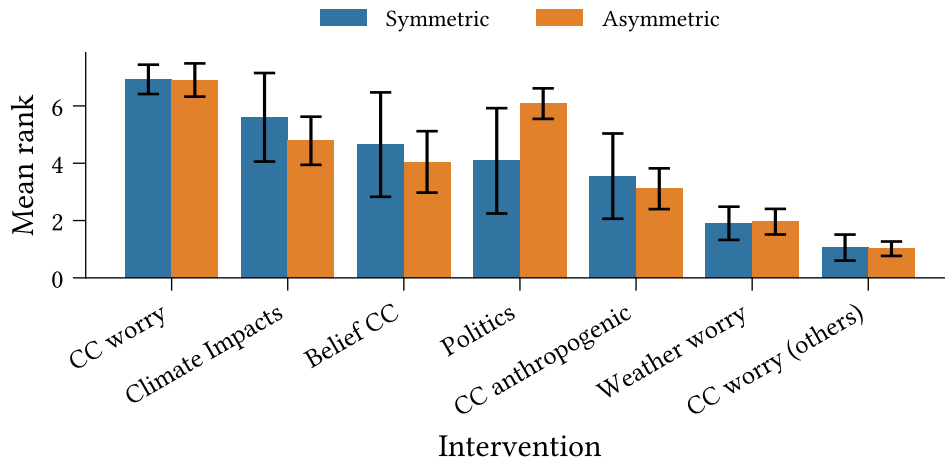


Figure 5.4: Expected ranking for weak interventions ($\delta h_i = 0.5$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

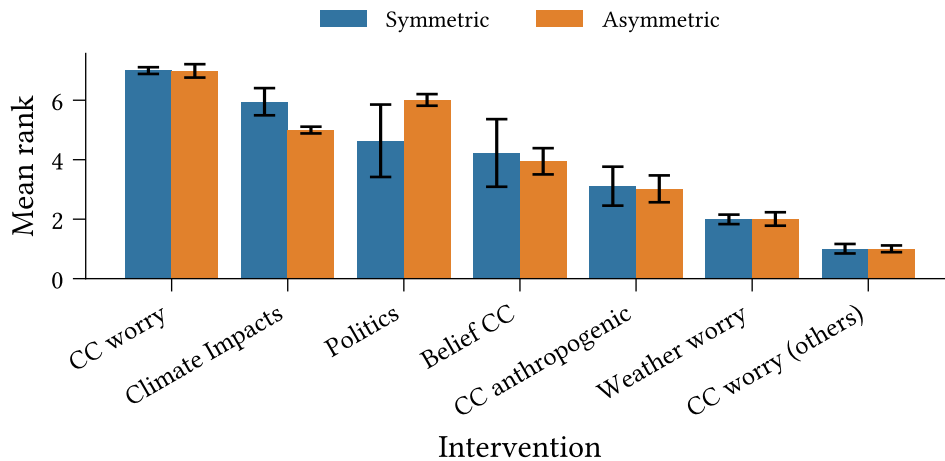


Figure 5.5: Expected ranking for medium interventions ($\delta h_i = 1.0$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

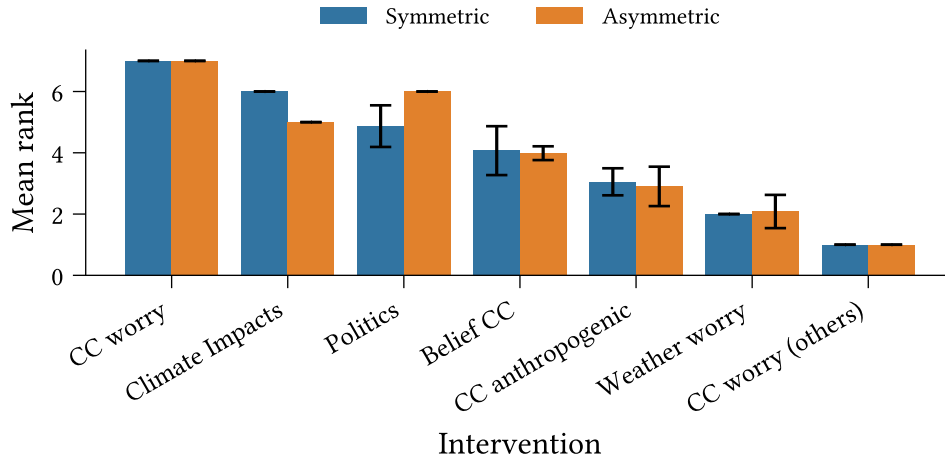


Figure 5.6: Expected ranking for perfect interventions ($\text{do}(S_i = +1)$) targeting individuals' support for climate-related policies. A higher rank indicates that an intervention is more effective, as measured by the proportion of the population who support climate policy after 30 months in simulation time.

5.4.2 Intervention effect

Measures the average difference in state between intervention and no-intervention scenarios

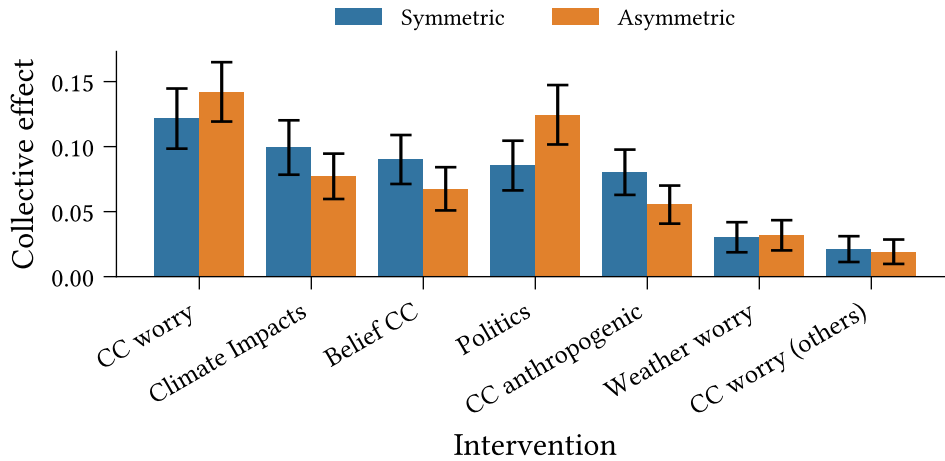


Figure 5.7: Expected effect of weak interventions ($\delta h_i = 0.5$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

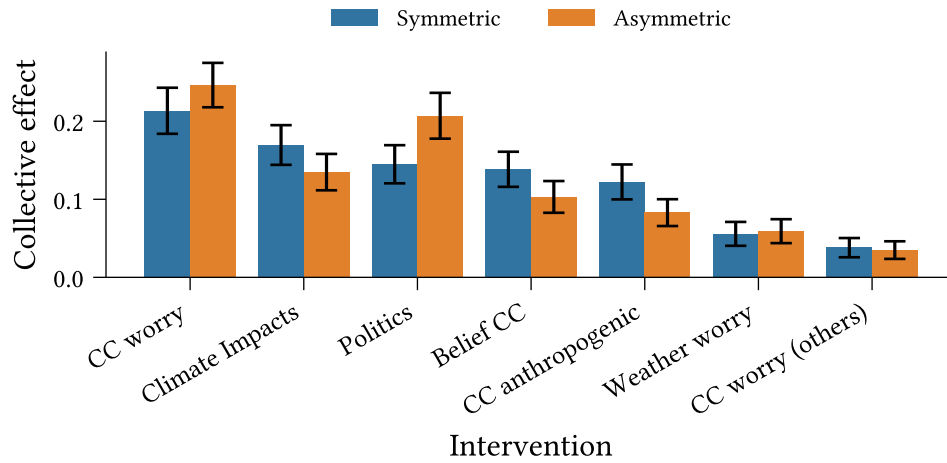


Figure 5.8: Expected effect of medium interventions ($\delta h_i = 1.0$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

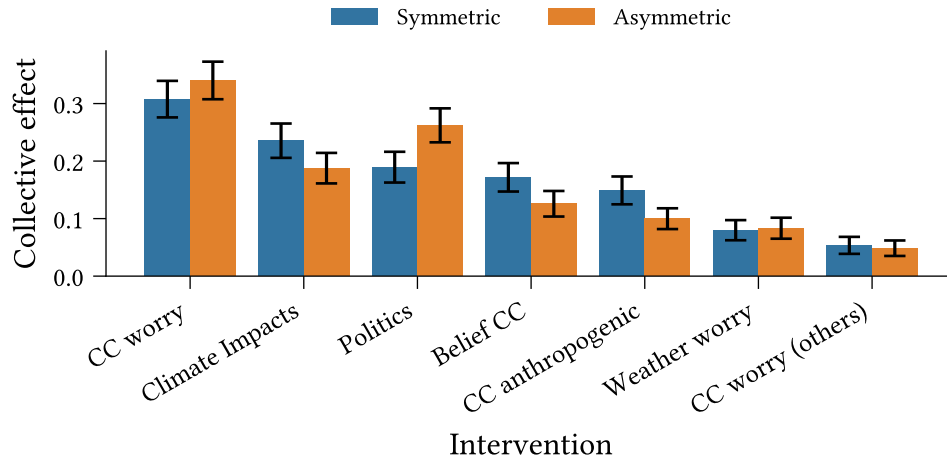


Figure 5.9: Expected effect of perfect interventions ($\text{do}(S_i = +1)$) targeting individuals' support for climate-related policies. Effect is calculated as the difference in state after 30 months, as observed in the intervention scenario, compared to the no-intervention scenario.

Chapter 6

Heterogeneous belief systems and intervention effects

6.1 Plan

Second results section, on individual heterogeneity in belief system structure, and intervention dynamics.

- **RQ2.2:** Fit models to conservative and liberal subsets of the data; compare and contrast. Compare with the model from the previous section, i.e., fit on the entire dataset.
- **RQ3.2:**
 - Distribution of intervention effectiveness by individual.
 - How does intervention ranking vary across individuals?
 - Characterising how different initial states affect intervention success.
 - Looking at how other theory-driven features affect success, e.g.:
 - How receptive is the individual to the intervention?

The effective baseline ($h_i + \sum_j J_{ji}s_j$) determines the probability that $S_i^{t+1} = s$ for a state $s \in \pm 1$. Evaluating this after intervening provides a measurement for the success of the intervention on the intervention spin itself.

We could look at how this changes with different intervention strengths (it follows a logistic curve).

- How consistent is the individual's belief state, as measured by the total system energy?

- How ‘entrenched’ is the target attitude?

Measure $h_k s_k + \sum_j J_{jk} s_j s_k$, where k is the target attitude.

6.2 RQ2.2: differences between belief systems for different (types of) individuals

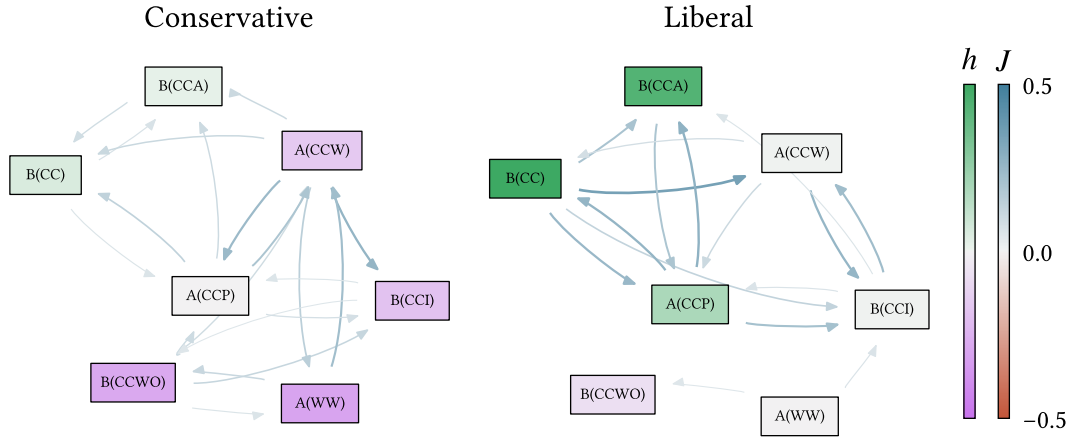


Figure 6.1: Prefixes: A (attitude), B (belief); node labels: CC (climate change), CCA (climate change anthropogenic), CCW (climate change worry), CCWO (climate change worry others), CCI (climate change impacts), CCP (climate change policies), WW (weather worry).

6.3 RQ3.2: Intervention strategy and effectiveness across individuals

Examine distributions of expected outcome effects (per individual) for different interventions. Characterise the individuals found at different parts of the distribution (personas, average magnetisation).

6.3.1 Intervention ranking

For each individual, compute ranking over the expected effects of each possible intervention.

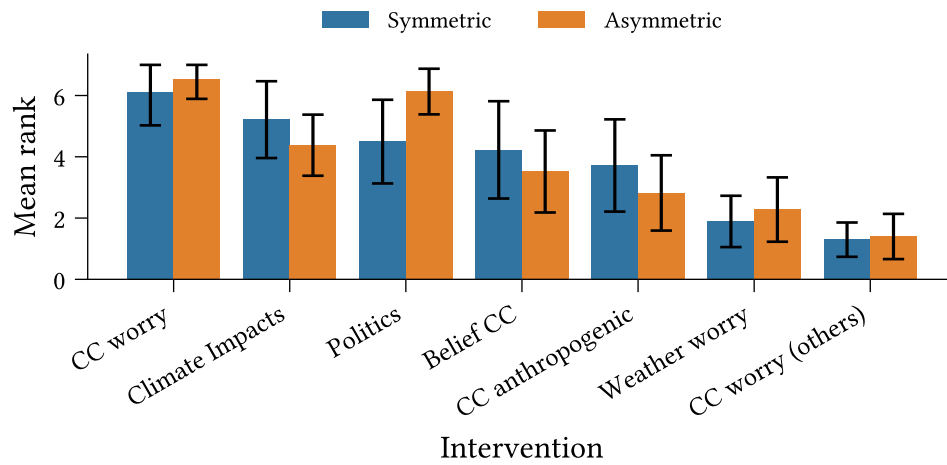


Figure 6.2: Ranked intervention effects per-individual, weak, targeting climate policy

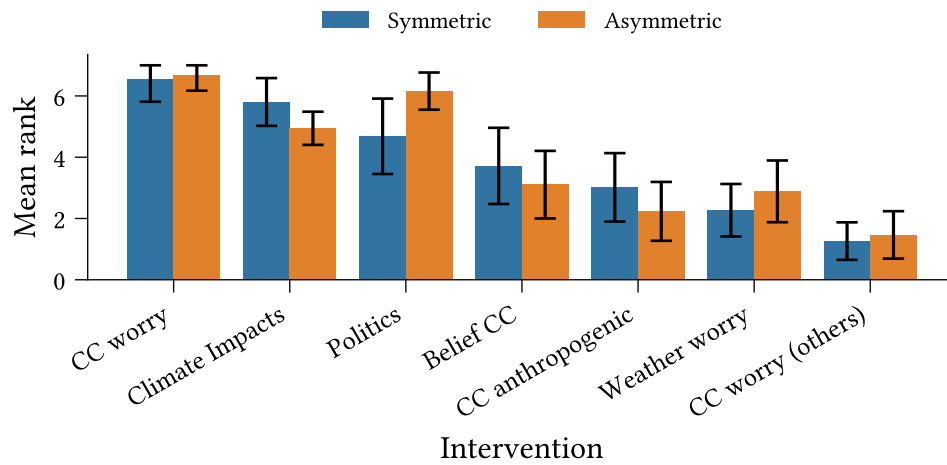


Figure 6.3: Ranked intervention effects per-individual, perfect, targeting climate policy

Chapter 7

Discussion

7.1 Plan

- Draw out the key findings from the above two results sections
- Make clear the implications for our theory of belief system dynamics
- Sensitivity of parameter estimation to unmeasured factors or incorrect structural assumptions. i.e., what happens when we cannot include an influential belief, or when we falsely assume that a relation does or does not exist?
- Representational limitations of our model:
 - Pairwise relations limit what can be modelled. Demonstrate using vaccination example — we can't capture relations between pairs of attitudes, whose effect sign or magnitude depends on a third belief/attitude.
 - Polar $(-1, +1)$ spin state assumption; some beliefs may be better treated as 'on/off', where the 'off' state has *no* effect on other spins, rather than a negative effect.
 - More generally, effect sizes may vary depending on specific spin states.

Chapter 8

Climate beliefs dataset

8.1 Plan

- Introduce the (insert name here) dataset, give context, survey details
- Validation, cleaning, transformations
- Question selection, indexes
- Binarisation

8.2 Longitudinal climate attitudes survey

Introduce the survey:

- Motivate it: This is the main resource used to construct our dataset.
- Description:
 - Content: What sorts of questions? (Examples)
 - Waves: how many, when were they run? (Participation event plot)
 - Participation: counts, variability. (Statistics: how many people answered all waves? How many singletons?)
 - Question variability. (...?)
- Overlap with notable events:
 - COVID-19 outbreaks
 - US electoral cycle
 - Major weather events

The longitudinal study (**CITE**) comprises six waves of responses from individuals residing in the United States, collected between April 2020 and March 2023. The assessed dimensions include general demographic information (e.g., age, gender, education, and

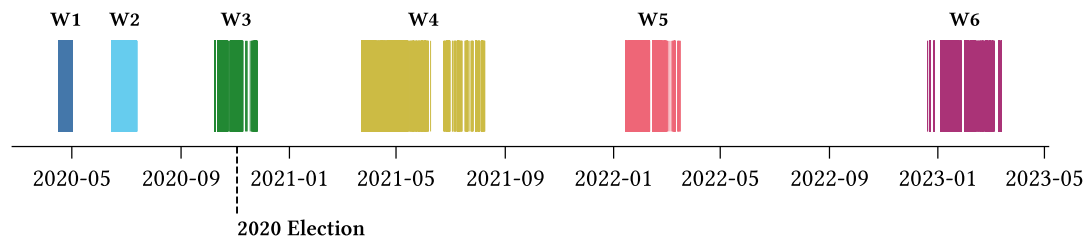


Figure 8.1: Longitudinal survey response dates per-wave

financial status), beliefs, attitudes, and experiences relating to concurrently-salient topics such as COVID-19, climate change, or the 2020 US presidential election, and support for hypothetical policies.

- Significance of longitudinal data — most studies use cross-sectional data.

Survey content; nature of the questions asked

- Context
- Category (belief, attitude, demographic, etc.)
- Response formats

The survey content varies between waves, and between participants.

Participation

Waves The survey waves occur with irregular spacing and duration, as illustrated in Figure 8.1. Some consecutive pairs of waves are much closer than others. For instance, the joined interval spanned by the first three waves has a longer duration than the interval between Waves 5 and 6, and the duration spanned by Wave 4 exceeds that of the joined interval comprising Waves 1 and 2.

The 2020 US Presidential Election date occurred during Wave 3. This wave was modified for responses started after the election date; future-focused survey questions regarding individuals' voting intentions, beliefs, and attitudes concerning the election were removed.

- Context of the survey:
 - Time-span
 - Wave timing/duration
 - Overlap with COVID-19 outbreaks and US electoral cycle, major weather events
- Codebook covers first five waves
- Nature of the questions asked:
- Variability in questions (between waves, participants)

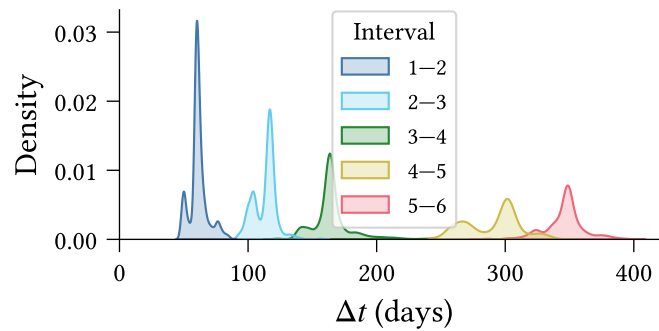


Figure 8.2: Between-response time distribution

8.3 Question selection

- Motivation (identify measures of distinct beliefs and attitudes regarding climate change – impacts, risks, policies, etc.)
- Criteria for selecting questions
- Exclude sixth wave due to no codebook

8.4 Validation

- Only validate selected questions
- Validation components:
 - Schema (Pandera)
 - Null responses iff null expected
 - Valid survey responses
 - Miscellaneous inconsistencies
 - Born in US in some waves, not in others
 - Changing responses to extreme weather experience in last 10 years
- Results:
 - Schema good
 - Various issues with Null responses:
 - Some due to survey logic errors
 - Some not explained

8.5 Normalisation, cleaning, and transformations

- Variable names (replace ‘ ’ with double underscore)
- Filter out participants with:
 - Null `participant_id`

- Survey errors (from validation above)
- Failed survey validation checks (if any check is failed, participant is removed entirely)
- Response type/value conversions
 - Non-float numeric variables to Int64
 - Re-map $1.. = k$ variables to $(0.. = k - 1)$
 - Replace empty string entries with `Null`
 - Multiple choice questions: comma-separated string of integers to list of integers/enums
 - Replace categorical variable integer responses with enums (for human-readability and to ensure value consistency across waves)
 - cc2: Re-order such that human-causes are ‘high values’ and natural-causes/no climate change are ‘low values’
- Coalescing treatment columns:
 - Responses stored in one column
 - Separate index column specifies treatment group membership
 - Optional third column for treatment-specific numeric parameters (e.g., policy cost)
- Constructed columns:
 - Merge `pol_party` and `pol_lean` to create five-point `pol_affiliation` scale. Note that this does not capture individuals who ‘lean *right* toward Democrat’ for example.

8.6 Data Validation

The complexity of the climate attitudes survey (**reference section discussing this**) necessitates a rigorous approach to data validation, both to identify errors in the expected schema and to ensure that the data matches our expectations.

We have implemented a general validation pipeline comprising three stages:

1. **Type-level validation:** The set of columns is exactly as expected, and all columns have the correct data type.
2. **Response-value validation:** All *non-null* responses are valid according to the schema specified in the codebook.
3. **Null-value validation:** Responses are null if, and only if, we expect them to be null.

In the first instance, type-level validation ensures that the *observed* data schema matches the *prescribed* schema. Since response data types vary between questions according to

the question type (see Table 8.1), the type-checking must be flexible and capable of handling complex data types.

For instance, categorical multiple-choice responses are represented using a `list[Enum]`, where `Enum` is a question-specific enum-type¹. Type validation for a multiple-choice question thus requires checking that the response column comprises `list`'s whose elements are all from the set of values defined by the `Enum`.

Response-value validation then ensures that all non-null responses are valid according to the set of allowable values described in the survey codebook. For most variables this is straightforward. Numeric and single-response ordinal variables typically have a defined range (e.g., $18 \leq \text{age} \leq 99$, or $1 \leq x \leq 5$ for a 5-point Likert scale variable). Text-entry responses are always considered valid.

Single-response categorical questions have `Enum` type, so type-level validation is sufficient to ensure that the responses do not contain any values outside the set defined by the `Enum`. However, the allowable values occasionally change between survey waves. Hence response-value validation is also required for categorical single-response and multiple-response variables, and in general must handle with schema variation between waves.

We implement both type-level and response-value validation using the `Pandera` Python library for DataFrame validation (Bantilan et al., 2022). The `Pandera` API allows for straightforward schema specification using Python type annotations (for type-level validation) in combination with value constraints (for response-level validation). We implement a separate validation check for questions whose response schema varies between survey waves, using a `Pandera` ‘dataframe check’ to stratify validation by wave.

Null-value validation ensures that responses are null if, and only if, they are expected to be null. For a question Q , the response of a participant P in wave W can be null for one of four reasons:

Response schema	Raw type	Coerced type
Text	<code>str</code>	<code>str</code>
Single response (ordinal)	<code>int</code>	<code>int</code>
Single response (categorical)	<code>int</code>	<code>Enum</code>
Multiple response	<code>str</code>	<code>list[Enum]</code>
Numeric	<code>float</code>	<code>float int</code>

Table 8.1: **TODO**

¹An enum is a type defined by a finite set of allowable values. In our case the values are human-readable strings. For instance, the `dem_urban` survey question, which asks ‘What kind of area do you live in?’ has responses with data type described by the enum `{Urban, Suburban, Rural}`.

- N1. Q is not asked in wave W ,
- N2. In wave W , Q is only shown to *new* participants, but P is a *repeating* participant (or vice versa),
- N3. In wave W , Q is only shown to a treatment group of which P is not a member, or
- N4. In wave W , Q is conditionally shown to participants based on their response to a distinct question Q' , and P 's response to Q' does not satisfy the condition.

Formally we can write the condition for P 's response R being null as:

$$\text{null}(R) \iff (\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4}) \quad (8.1)$$

We split this up into: If not in wave, then R is null.

$$(\text{N1} \vee \text{N2}) \implies \text{null}(R) \quad (8.2)$$

If conditions not met, then R is null.

$$(\text{N3} \vee \text{N4}) \implies \text{null}(R) \quad (8.3)$$

For unconditional columns, if shown, then not null:

$$\neg(\text{N1} \vee \text{N2}) \implies \neg \text{null}(R) \quad (8.4)$$

For conditional columns, if shown, then not null:

$$\neg(\text{N1} \vee \text{N2} \vee \text{N3} \vee \text{N4}) \implies \neg \text{null}(R) \quad (8.5)$$

For a conditional $A \implies B$, let $\mathbf{R}$ be the subset of responses for which the contradiction $\neg(A \implies B) \equiv (A \wedge \neg B)$ is true. Then $A \implies B$ is satisfied if, and only if, $\mathbf{R}$ is empty **check this — is it iff, if, or only if?**. If $\mathbf{R}$ is non-empty, then the contained responses are counter-examples to the conditional.

- Single-response ordinal:

In the first case, type-level validation ensures that the data schema matches expectations, independently of the response values. In the first case, type-level validation ensures that the schema observed in the dataset matches the schema Type-level validation Our validation pipeline comprises three stages:

The first two stages are handled by Pandera, and the third is manually implemented.

- What does type-level validation consist in?
 - Simple types
 - Enums
 - Lists of enums
- What does response value validation consist in?
 - Checking that observed (non-null) values are within the set of allowable values
- What does null-value validation consist in?
 - ...

The data schema defines everything that is necessary. Maybe we can discuss this separately as an implementation step.

Chapter 9

Related work

Considering moving this section to later in the thesis.

- Early work on modelling belief systems, triadic consistency
- Approaches:
 - partial correlation networks,
 - Bayesian networks,
 - Causal attitude network, Ising-style models
- Beliefs as edges vs. beliefs as nodes
- Social influence: hierarchical Ising models, network of belief model
- Within-person vs between-person correlations; complications around inferring individual belief systems.

Chapter 10

Conclusions and future work

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doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Chapter 11

Ethics and Data Management

A new requirement for the thesis is that there must be a short section in which you reflect on the ethical aspects of your project. This requirement is related to one of the final objectives that a graduated student of the Master of Computational Science must meet: “The graduate of the program has insight into the social significance of Computational Science and the responsibilities of experts in this field within science and in society“. You don’t need to devote an entire chapter to this; a short section or paragraph is sufficient.

I acknowledge that the thesis adheres to the ethical code (<https://student.uva.nl/en/topics/ethics-in-research>) and research data management policies (<https://rdm.uva.nl/en>) of UvA and IvI.

The following table lists the data used in this thesis (including source codes). I confirm that the list is complete and the listed data are sufficient to reproduce the results of the thesis. If a prohibitive non-disclosure agreement is in effect at the time of submission “NDA” is written under “Availability” and “License” for the concerned data items.

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